Class 10 Science: Chapter 1 — Chemical Reactions and Equations

1. Introduction to Chemical Reactions

A **chemical reaction** is a process in which one or more substances (**reactants**) undergo a chemical change to form new substances (**products**) with entirely different physical and chemical properties.

Key Observations Confirming a Chemical Reaction

A chemical reaction has taken place if one or more of the following observations occur:

1. **Change in state** (e.g., gas to liquid/solid)
2. **Change in colour**
3. **Evolution of a gas**
4. **Change in temperature** (release or absorption of heat)
5. **Formation of a precipitate** (an insoluble solid)

Core NCERT Practical Activities

- **Activity 1.1 — Burning of Magnesium Ribbon in Air**
  - **Observation**: Magnesium ribbon burns with a dazzling white flame to form a white powder of Magnesium Oxide (MgO).
  - **Exam Question**: *Why is a magnesium ribbon cleaned with sandpaper before burning?*
    - **Answer**: It is cleaned to remove the protective layer of basic magnesium carbonate formed on its surface due to slow reaction with moist air, allowing it to burn readily.
  - **Equation**:

$$2\mathrm{Mg}(s) + \mathrm{O}_2(g) \rightarrow 2\mathrm{MgO}(s)$$

- **Activity 1.2 — Reaction between Lead Nitrate and Potassium Iodide**
  - **Observation**: A bright yellow precipitate of Lead Iodide (PbI_2) is formed.
  - **Equation**:

$$\mathrm{Pb(NO_3)_2}(aq) + 2\mathrm{KI}(aq) \rightarrow \mathrm{PbI_2}(s) \downarrow + 2\mathrm{KNO_3}(aq)$$

- **Activity 1.3 — Action of Dilute Acid on Zinc Granules**
  - **Observation**: Effervescence (bubbles) of Hydrogen gas (H_2) appears around zinc granules. The conical flask becomes warm (exothermic reaction).
  - **Equation**:

$$\mathrm{Zn}(s) + \mathrm{H_2SO_4}(aq) \rightarrow \mathrm{ZnSO_4}(aq) + \mathrm{H_2}(g) \uparrow$$

2. Chemical Equations and Balancing

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A **chemical equation** is the symbolic representation of a chemical reaction using chemical formulas of the reactants and products.

Balanced Chemical Equation

A chemical equation in which the total number of atoms of each element on the reactant side equals the total number of atoms on the product side.

- **Law of Conservation of Mass**: Mass can neither be created nor destroyed in a chemical reaction. Therefore, the total mass of reactants must equal the total mass of products.

Making Equations More Informative

1. **Physical States**: Represented by symbols: Solid (s), Liquid (l), Gas (g), Aqueous solution (aq).
2. **Gas & Precipitate**: Evolved gas is indicated by ↑; precipitate formed is indicated by ↓.
3. **Reaction Conditions**: Temperature, pressure, or catalysts are written above or below the reaction arrow (→).
4. **Energy Changes**: +Heat on the product side denotes exothermic; +Heat on the reactant side (or $\xrightarrow{\Delta}$) denotes endothermic.

Step-by-Step Method to Balance Equations (Hit and Trial Method)

Example: Balance $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

1. **Write the skeletal equation**: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
2. **Count the number of atoms of each element**:
   - Left side (Reactants): $Fe = 1, H = 2, O = 1$
   - Right side (Products): $Fe = 3, H = 2, O = 4$
3. **Balance the element with maximum atoms first** (O): Multiply H_2O by 4:

$$Fe + 4H_2O \rightarrow Fe_3O_4 + H_2$$

4. **Balance Hydrogen** (H): Multiply H_2 on product side by 4:

$$Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$$

5. **Balance Iron** (Fe): Multiply Fe on reactant side by 3:

$$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$$

6. **Final Balanced Equation with Physical States**:

$$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$$

3. Types of Chemical Reactions

A. Combination Reaction

A reaction in which **two or more substances combine to form a single product**.

- **General Form**: $A + B \rightarrow C$

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- **Key Reactions**:
  1. **Slaking of Lime**: Quicklime (CaO) reacts vigorously with water releasing a large amount of heat to form Slaked lime ($Ca(OH)_2$).

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + \text{Heat}$$

     - **Exam Note (Whitewashing)**: Solution of slaked lime is applied to walls. It reacts slowly with CO_2 in air over 2–3 days to form a shiny layer of Calcium Carbonate ($CaCO_3$):

$$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$$

  2. **Burning of Coal**:

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

  3. **Formation of Water**:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

B. Decomposition Reaction

A reaction in which a **single compound breaks down into two or more simpler substances**. Heat, light, or electricity is required to break the bonds.

Types of Decomposition

1. **Thermal Decomposition** (by Heat):
   - **Heating Ferrous Sulphate crystals** ($FeSO_4 \cdot 7H_2O$): Green crystals lose water molecules turning white, then break down into reddish-brown Ferric oxide and suffocating fumes of sulfur gases.

$$2FeSO_4(s) \xrightarrow{\Delta} Fe_2O_3(s) + SO_2(g) + SO_3(g)$$

   - **Heating Lead Nitrate**: Emits brown fumes of Nitrogen Dioxide (NO_2) leaving a yellow residue of Lead Oxide (PbO).

$$2Pb(NO_3)_2(s) \xrightarrow{\Delta} 2PbO(s) + 4NO_2(g) \uparrow + O_2(g) \uparrow$$

   - **Decomposition of Calcium Carbonate**: Used in cement manufacturing.

$$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$$

2. **Electrolytic Decomposition (Electrolysis)** (by Electricity):
   - **Electrolysis of Water**: Water breaks into hydrogen and oxygen gas.

$$2H_2O(l) \xrightarrow{\text{Electricity}} 2H_2(g) + O_2(g)$$

   - **Exam Key Point**: Volume of H_2 gas collected at the **Cathode** (negative electrode) is **twice** the volume of O_2 gas collected at the **Anode** (positive electrode) because water contains hydrogen and oxygen in a 2:1 ratio by volume.
3. **Photolytic Decomposition** (by Light):

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- **Decomposition of Silver Chloride**: White AgCl turns grey in sunlight.

$$2AgCl(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Cl_2(g)$$

- **Decomposition of Silver Bromide**:

$$2AgBr(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Br_2(g)$$

- **Application**: Used in **Black and White Photography**.

C. Displacement Reaction

A chemical reaction in which a **more reactive element displaces a less reactive element** from its salt solution.

- **General Form**: $A + BC \rightarrow AC + B$
- **Key Reactions**:

1. **Iron nail in Copper Sulphate solution**:
   - **Observation**: Blue colour of $CuSO_4$ fades to light green ($FeSO_4$), and reddish-brown copper deposits on iron nail.

$$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$$

2. **Zinc in Copper Sulphate solution**: Blue solution becomes colourless.

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

3. **Lead in Copper Chloride solution**:

$$Pb(s) + CuCl_2(aq) \rightarrow PbCl_2(aq) + Cu(s)$$

D. Double Displacement Reaction

A reaction in which there is an **exchange of ions between reactants** to form two new compounds.

- **General Form**: $AB + CD \rightarrow AD + CB$
- **Precipitation Reaction**: Any double displacement reaction that produces an insoluble solid precipitate.
- **Key Reactions**:

1. **Sodium Sulphate and Barium Chloride**:
   - **Observation**: White precipitate of Barium Sulphate ($BaSO_4$) forms immediately.

$$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) \downarrow + 2NaCl(aq)$$

2. **Lead Nitrate and Potassium Iodide**:

$$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) \downarrow + 2KNO_3(aq)$$

4. Classification Based on Energy / Heat Flow

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Property	Exothermic Reactions	Endothermic Reactions
Definition	Reactions in which heat energy is **released** along with products.	Reactions in which heat energy is **absorbed** from surroundings.
Temperature	Temperature of reaction system increases.	Temperature of reaction system decreases.
Examples	• Respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$ • Burning of Natural Gas ($CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$) • Decomposition of vegetable matter into compost	• All decomposition reactions (Thermal, Electrolytic, Photolytic) • Photosynthesis: $6CO_2 + 6H_2O \xrightarrow{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$

5. Oxidation and Reduction Reactions (Redox Reactions)

Definitions

- **Oxidation**: Addition of Oxygen **OR** Removal of Hydrogen from a substance.
- **Reduction**: Addition of Hydrogen **OR** Removal of Oxygen from a substance.
- **Redox Reaction**: A reaction in which oxidation and reduction occur simultaneously.

Agent Identification Rules

- **Oxidizing Agent**: The reactant that gives oxygen or removes hydrogen (it gets reduced itself).
- **Reducing Agent**: The reactant that gives hydrogen or removes oxygen (it gets oxidized itself).

Exemplar Redox Analysis

Case 1: Reaction of Copper Oxide with Hydrogen

$$CuO(s) + H_2(g) \xrightarrow{\Delta} Cu(s) + H_2O(g)$$

- **Substance Oxidized**: H_2 (Gained oxygen → H_2O)
- **Substance Reduced**: CuO (Lost oxygen → Cu)
- **Oxidizing Agent**: CuO

- **Reducing Agent**: H_2

Case 2: Reaction of Manganese Dioxide with Hydrochloric Acid

$$MnO_2(s) + 4HCl(aq) \rightarrow MnCl_2(aq) + 2H_2O(l) + Cl_2(g)$$

- **Substance Oxidized**: HCl (Lost hydrogen → Cl_2)
- **Substance Reduced**: MnO_2 (Lost oxygen → $MnCl_2$)
- **Oxidizing Agent**: MnO_2
- **Reducing Agent**: HCl

Case 3: Reaction of Zinc Oxide with Carbon

$$ZnO(s) + C(s) \rightarrow Zn(s) + CO(g)$$

- **Substance Oxidized**: C → CO
- **Substance Reduced**: ZnO → Zn
- **Oxidizing Agent**: ZnO
- **Reducing Agent**: C

6. Effects of Oxidation in Everyday Life

A. Corrosion

The process in which metals are slowly eaten away by the action of air, moisture, acids, or chemicals on their surface.

- **Key Examples**:
    1. **Rusting of Iron**: Hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$) formed as reddish-brown flaky substance.
    2. **Tarnishing of Silver**: Silver turns black due to formation of Silver Sulphide (Ag_2S).
    3. **Corrosion of Copper**: Copper acquires a green coating of basic copper carbonate ($CuCO_3 \cdot Cu(OH)_2$).
- **Prevention Methods**:
    - Painting, oiling, greasing.
    - **Galvanization**: Process of applying a protective thin coating of **Zinc** on iron/steel.
    - Electroplating, chrome plating, and making alloys (e.g., Stainless steel).

B. Rancidity

The oxidation of fats and oils present in food products when exposed to air, leading to bad smell, unpleasant taste, and spoilage.

- **Prevention Methods**:

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1. Adding **Antioxidants** to food products containing oils and fats.
2. **Flushing containers/bags with inert gases** like Nitrogen (N_2) (e.g., potato chip bags).
3. Storing food in **airtight containers**.
4. Keeping food in **refrigerators** (low temperature slows down oxidation).
5. Storing food away from direct sunlight.

7. High-Yield Board Exam Formula & Colour Guide

Compound Name	**Formula**	**Chemical Name**	**Characteristic Colour / State**
Quicklime	CaO	Calcium Oxide	White solid
Slaked lime	$Ca(OH)_2$	Calcium Hydroxide	Clear/milky solution
Limestone / Marble	$CaCO_3$	Calcium Carbonate	White solid
Ferrous Sulphate crystals	$FeSO_4 \cdot 7H_2O$	Ferrous Sulphate Heptahydrate	Green crystals
Ferric Oxide	Fe_2O_3	Iron(III) Oxide	Reddish-brown solid
Lead Oxide	PbO	Lead(II) Oxide	Yellow solid residue
Nitrogen Dioxide	NO_2	Nitrogen Dioxide	Brown fumes
Copper Powder	Cu	Copper metal	Reddish-brown solid
Copper Oxide	CuO	Copper(II) Oxide	Black powder
Copper Sulphate solution	$CuSO_4(aq)$	Copper Sulphate	Blue solution
Ferrous Sulphate solution	$FeSO_4(aq)$	Ferrous Sulphate	Light green solution

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Compound Name	Formula	Chemical Name	Characteristic Colour / State
Lead Iodide	PbI_2	Lead(II) Iodide	Bright yellow precipitate
Barium Sulphate	$BaSO_4$	Barium Sulphate	White precipitate
Silver Chloride	AgCl	Silver Chloride	White solid (turns grey in light)

8. Exam-Focused High-Yield Questions

Q1: Why is respiration considered an exothermic reaction?

Answer: Respiration is the process of breaking down glucose ($C_6H_{12}O_6$) by combining it with oxygen in living cells to produce energy along with CO_2 and H_2O. Because heat energy is liberated in this process, respiration is exothermic:

$$C_6H_{12}O_6(aq) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l) + \text{Energy}$$

Q2: A shiny brown element 'X' on heating in air becomes black in colour. Name element 'X' and the black colored compound formed.

Answer: Element 'X' is **Copper** (Cu). The black colored compound formed is **Copper(II) Oxide (CuO)**.

$$2Cu(s) + O_2(g) \xrightarrow{\Delta} 2CuO(s)$$

Q3: What happens when potassium iodide solution is added to lead nitrate solution? Write the chemical equation and state the type of reaction.

Answer: A bright yellow precipitate of Lead Iodide (PbI_2) is formed. This is a **Double Displacement (Precipitation) reaction**.

$$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) \downarrow + 2KNO_3(aq)$$

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